

SMICLR: Contrastive Learning on Multiple Molecular Representations for Semisupervised and Unsupervised Representation Learning

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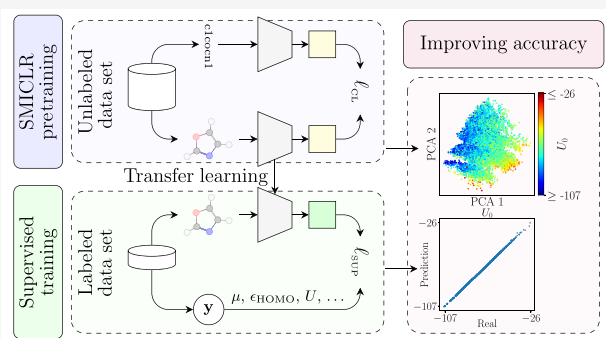


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ABSTRACT: Machine learning as a tool for chemical space exploration broadens horizons to work with known and unknown molecules. At its core lies molecular representation, an essential key to improve learning about structure–property relationships. Recently, contrastive frameworks have been showing impressive results for representation learning in diverse domains. Therefore, this paper proposes a contrastive framework that embraces multimodal molecular data. Specifically, our approach jointly trains a graph encoder and an encoder for the simplified molecular-input line-entry system (SMILES) string to perform the contrastive learning objective. Since SMILES is the basis of our method, i.e., we built the molecular graph from the SMILES, we call our framework as SMILES Contrastive Learning (SMICLR). When stacking a nonlinear regressor on the SMICLR’s pretrained encoder and fine-tuning the entire model, we reduced the prediction error by, on average, 44% and 25% for the energetic and electronic properties of the QM9 data set, respectively, over the supervised baseline. We further improved our framework’s performance when applying data augmentations in each molecular-input representation. Moreover, SMICLR demonstrated competitive representation learning results in an unsupervised setting.



1. INTRODUCTION

In the last years, the progress in self-supervised learning for representation learning has attracted a lot of attention.^{1–3} Roughly, the core idea of this paradigm is to use information from the unlabeled data itself as the supervision signal when training a machine learning (ML) model. One of the mainstream self-supervised approaches is contrastive learning (CL), which has already demonstrated profound success in diverse applications, such as computer vision,^{4,5} natural language processing,^{6,7} speech processing,⁸ and graph representation learning.^{9–11} Thus, there are great expectations that those techniques can improve ML predictions of molecular properties.

In contrastive learning, data are composed of similar (or positive) and dissimilar (or negative) pairs. For self-supervised algorithms, the acquisition of positive examples generally involves different augmented views of the same data (e.g., data augmentation), whereas negative pairs are usually the other samples in the mini-batch. The objective function is to learn an embedding space that draws nearby distance representations for positive pairs and far-away representations for negative pairs.^{2,5}

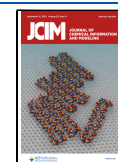
For graph-structured data, efforts have been made to use CL as a pretrain task in an end-to-end framework that involves graph neural networks (GNNs). For instance, Sun et al.¹⁰ proposed the InfoGraph, a GNN-based method that maximizes the mutual information (MI) between the global- and patch-level

representations to capture aspects shared across the graph substructures. You et al.¹¹ introduced a graph contrastive learning (GraphCL) framework that learns an embedding space by maximizing agreement between correlated views of the same graph. For this, their work investigated several graph augmentation strategies (e.g., node dropping, edge perturbation) and showed that the composition of various augmentations could further enhance CL performance. More recently, Hassani and Khasahmadi combined both previous strategies as a framework that maximizes the MI between graph- and node-level representations on augmentation views of the same graph.¹²

A graph can naturally encode a molecule by representing the atoms and chemical bonds as nodes and edges, respectively.^{13,14} In the literature, GNN algorithms have made remarkable improvements for molecular representation learning.^{15–17} On this basis, Gilmer et al.¹⁸ proposed the Message Passing Neural Networks (MPNN), a framework that abstracts several GNN

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models. Later, Klicpera et al.¹⁹ extends the MPNN by incorporating directional information such as bond angles and rotations. Schütt et al.²⁰ presented a GNN architecture with continuous-filter convolutional layers for modeling quantum interactions in molecules.

Nevertheless, a few studies attempted to harness GNN with unlabeled data to enhance chemical space learning. In addition to the before-mentioned CL approaches,^{10–12} Wang and colleagues²¹ designed a contrastive framework for molecules, called MolCLR, by combining GNN-based encoders and graph augmentations. Under these circumstances, the recent results already suggest that additional information about the chemical space increases accuracy. Thus, from our understanding, the representation learned by CL algorithms may be a pivotal piece to improve ML models for molecular property prediction and materials discovery.

Machine learning studies have used a wide range of molecular representations,²² for example, images,^{23,24} string-based representations,^{25,26} molecular descriptors,^{27,28} fingerprints,²⁹ and parse trees.^{30,31} In this realm, the FragNet³² is a CL approach that exploits the simplified molecular-input line-entry system (SMILES) to represent the molecular structure and the SMILES enumeration as the data augmentation technique. Moreover, since molecular representations encode a specific molecule aspect (e.g., 2D and or 3D information), contrastive models may benefit from these multiple data domains. To investigate this point, Sanchez-Fernandez et al.³³ introduced the CLOOME, a multimodal CL framework trained on the microscopy data and Morgan fingerprints.

In this work, we propose a multimodal CL framework, similar to the idea presented by Sanchez-Fernandez et al.,³³ however, leveraging two popular molecular representations and augmentations for each modality. To do so, we exploit two molecular representations that can be easily acquired from chemical space: (i) the SMILES string and (ii) the molecular graph. The SMILES can be produced by describing the molecule based on basic syntax rules, whereas the graph can be obtained through the conversion of the SMILES. In order to deal with these distinct inputs, we jointly trained a graph and text encoder to learn the latent representation. Moreover, we explore various augmentations as using it demonstrated to be an essential prerequisite for CL.^{5,11}

As a result, we show that our framework, named SMICLR, which stands for **SMILES Contrastive Learning**, achieved state-of-the-art performance in unsupervised representation learning for a graph classification downstream task and property prediction in a semisupervised setting. When fine-tuning the base encoder to predict molecular properties from the QM9 data set,³⁴ on average, our approach reduced the prediction error by 35% over the supervised baseline. Moreover, we empirically demonstrated that SMICLR exhibits stunning performances when incorporating geometric information obtained from a low-cost method, thus opening opportunities to work with other state-of-the-art geometry-based GNNs.

In summary, our main contributions are listed below:

- We propose SMICLR, a CL framework that learns representation from multiple molecular data domains.
- We design and investigate diverse data augmentation techniques for the molecular graph and SMILES string representation.
- We benchmark the SMICLR semisupervised performance using 12 molecular properties of the QM9 data set³⁴

and evaluate the unsupervised representation learning in the MUTAG data set^{35,36} for a graph classification task using a support vector machine (SVM) classifier.

2. BACKGROUND

In this section, we detail the methods adopted by our proposed framework. Figure 1 summarizes the pipeline used to perform

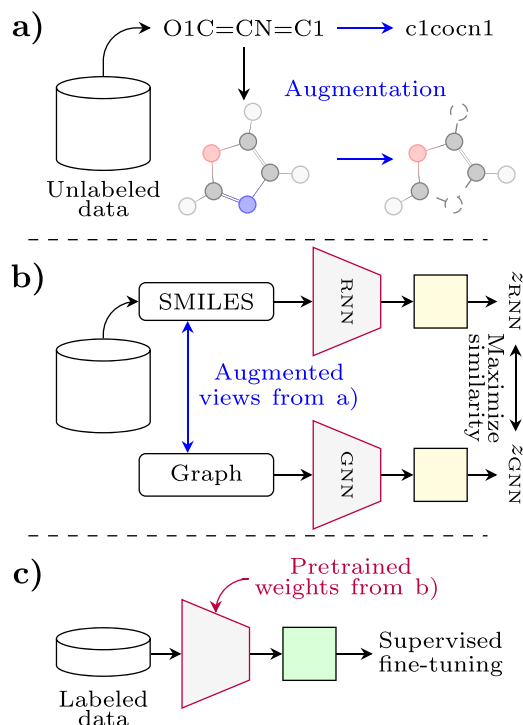


Figure 1. Overview of the SMICLR framework: from the pretext task to the downstream task.

the contrastive and prediction task. Starting with the SMILES of a particular molecule, we build its respective molecular graph representation. Next, we apply data augmentation techniques in the molecular representations (i.e., the SMILES and molecular graph). Then, we feed the encoders with the augmented views so that our framework can learn a latent space while performing CL. With the pretext task finished, we fine-tune our pretrained base encoder to predict molecular properties.

2.1. Molecular Representation. Two molecular representations commonly used are the SMILES string and the molecular graph. These representations rely on trivial information about the chemical compound, such as the atoms and chemical bonds in the molecule.²² In the literature, the SMILES is generally used as input to recurrent neural networks (RNNs), whereas the graph-structured data are directly fed into GNNs.^{18,37} Our framework exploits both representations as correlated views for CL, and consequently, it explores the before-mentioned neural network (NN) algorithms. In the following sections, we provide an overview of the related works.

2.2. Recurrent Neural Network. RNN is a generalization of a feed-forward NN that propagates its hidden state through sequential data to perform learning.³⁸ Hence, many works employed it with a SMILES string to tackle the problem of *de novo* molecular generation, typically by using long short-term memory (LSTM) cells. In this context, Gómez-Bombarelli et

al.³⁹ employed a variational autoencoder (VAE) with a multilayered RNN-based decoder to learn a continuous representation of the molecule. Similarly, Kusner et al.³⁰ used a parse tree to express the semantic constraints of the SMILES and improved the generation of valid molecules produced by the VAE model. Kotsias et al.⁴⁰ trained an RNN conditioned with molecular descriptors. Other RNN-based models for molecule generation have been covered in the detailed overview given by Elton and collaborators.³⁷

Specifically, an RNN is a specialized NN that learns sequence, or temporal, dependencies. Consider a sequence S composed of symbols $s_1, s_2, \dots, s_w$ (e.g., SMILES string), where $|S| = w$.⁴¹ At its basis, an RNN computes a hidden state h_t for each symbol $s_t \in S$, which corresponds to the output of a function $F(\cdot)$ that receives the current s_t and the previous hidden state h_{t-1} of the model (i.e., the hidden state from the sequence member that precedes s_t). Here, $F(\cdot)$ is a differentiable function that learns the most salient information through the sequence. The $F(\cdot)$ function varies according to the RNN layer, and c_{t-1} is the cell state, which is the long-term memory that composes only LSTM models.^{38,41} For the LSTM layer, $F(\cdot)$ is defined as follows:

$$i_t = \sigma(W_{si}s_t + b_{ii} + W_{hi}h_{t-1} + b_{hi}) \quad (1)$$

$$f_t = \sigma(W_{sf}s_t + b_{if} + W_{hf}h_{t-1} + b_{hf}) \quad (2)$$

$$g_t = \tanh(W_{sg}s_t + b_{ig} + W_{hg}h_{t-1} + b_{hg}) \quad (3)$$

$$o_t = \sigma(W_{so}s_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \quad (4)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \quad (5)$$

$$h_t = o_t \odot \tanh(c_t) \quad (6)$$

where the W and b terms are, respectively, the weight matrices and the bias, in which the subscript indices indicate the dimension, and $\odot$ is the Hadamard product. Here, σ and $\tanh$ are, respectively, the logistic sigmoid and hyperbolic tangent activation function. In order to determine the relevant information from s_t , c_{t-1} , and c_t , the LSTM counts with three gates, namely, the input gate i_t , the forget gate f_t , and the output gate o_t . Each gate comprises the σ function, which outputs numbers between zero and one to determine the informative component portion.³⁸

2.3. Graph Neural Network. GNN has gained popularity due to its state-of-the-art performances in diverse graph tasks, including molecular prediction tasks.^{20,42} It tries to encode the graph structure by an iterative message passing process. A wide variety of GNNs has been introduced recently.^{18–20} Among them, some GNN variants aimed to work with molecules or crystals data, such as the Crystal Graph Convolutional Neural Network and the Materials Graph Network.⁴³ In summary, GNN uses a two-phase iterative process to capture the representation of a graph: (i) First, it learns the node structural information by operating a neighborhood aggregation strategy called the message-passing scheme. (ii) Then, it uses a readout function to summarize the local representations into a graph level.⁴⁴

Consider an undirected graph $G=\{V,E\}$ that describes a molecule, where V and E denote the set of vertices (atoms) and edges (chemical bonds), respectively. Let x_v be the feature vector of a vertex $v \in V$ and e_{vw} be the attribute vector of an edge $e \in E$ between v and w , where $w \in V$. Given the t th layer of a GNN, the forward message passing operates as follows:

$$h_v^{t+1} = U_t \left(h_v^t, \sum_{w \in N(v)} M_t(h_v^t, h_w^t, e_{vw}) \right) \quad (7)$$

where h_v^{t+1} is the embedding vector of v at the $t+1$ th layer (note that $h_v^0 = x_v$). Here, $N(v)$ are neighbor vertices of v , and $M_t(\cdot)$ is the message function that collects information about the adjacent vertices and $U_t(\cdot)$ the vertex update function. After t stages, or t th layers, of message passing operations, the readout phase applies an $R(\cdot)$ function, as shown in eq 8, to summarize the set of hidden state h_v^t into a feature vector for the entire graph, h_G .

$$h_G = R(\{h_v^t | v \in \mathcal{G}\}) \quad (8)$$

Note that $M_t(\cdot)$, $U_t(\cdot)$, and $R(\cdot)$ are functions with learnable parameters that assume different forms depending on the GNN. Here, we adopt the enn-s2s model from Gilmer et al.¹⁸ for the semisupervised regression tasks and the graph isomorphism network (GIN) for unsupervised representation learning for a classification problem.⁴⁴

2.4. Semisupervised Learning. Besides the vast chemical space, estimated between the order from 10^{60} to 10^{100} chemical structures, a small fraction of molecules is known.⁴⁵ Semisupervised learning (SSL) transpires when only a tiny part of the samples in the data set contains their labels, and the remainder does not. Hence, the SSL may play an essential role here. In this scenario, Nguyen et al.⁴⁶ claims to be the first study that introduced an MPNN for molecular properties prediction in the SSL scenario. Zhang et al.⁴⁷ and Kang and Cho⁴⁸ proposed an autoencoder-based model with a semisupervised loss. Among these works, it is worth mentioning the methods that rely on language representation models, such as the transformer-based models.^{49,50} Recently, contrastive models have gained attention in SSL applications, as described in Section 2.5.

2.5. Contrastive Learning. Contrastive models try to draw close representations for similar samples and distant representations otherwise. As an unsupervised algorithm, CL treats each sample as its own class and defines the label annotation from the unlabeled data itself; i.e., it is a self-supervised method. When applied to an SSL setting, a common technique is to train the contrastive model on the entire data set and fine-tune its base encoder by adding a multilayer perceptron (MLP) to predict the desired output of the labeled set. Recent studies on CL have investigated the impact of their framework for molecular data in an SSL setting, downstream tasks, and transfer learning.^{10–12,21,32,33}

3. METHODOLOGY

The recent successful application of CL has shown its capability to learn valuable representations with minor human interaction.^{5,10,21} Here, we describe the CL framework proposed in this work and the augmentations adopted to create correlated views of the same molecule.

3.1. SMICLR Framework. Inspired by Chen et al.,⁵ we propose a contrastive framework that tries to learn a latent representation using two sources of augmentation view: (i) different molecular representations and (ii) transformations on the molecular representations. For this, we first generate the molecular graph given the SMILES string. The SMILES is the basis of the entire framework since we focus on a straightforward representation of unlabeled data. The graph representation follows the natural encoding; i.e., the atoms and chemical bonds are the graph's vertices and edges, respectively. Moreover, each

node and edge contains a set of features computed by cheminformatics libraries, such as RDKit,⁵¹ and OpenBabel,⁵² as we describe in Section 4.2. In the following, we detail every component of our framework. We also illustrate the flow of data of each module in Figure 2.

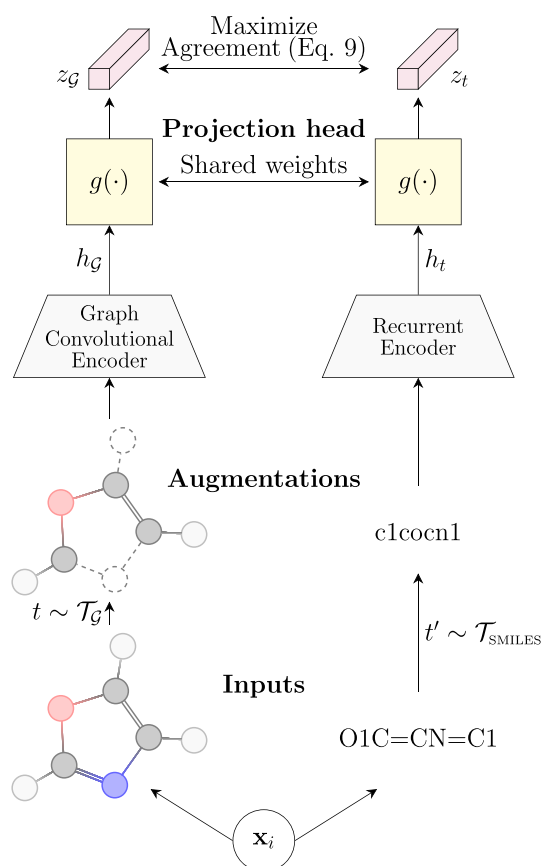


Figure 2. Overview of the SMICLR framework. Given the SMILES of a sample $\mathbf{x}_i$, it first obtains the graph representation. Then, two data augmentation operators T_G and T_{SMILES} are applied, respectively, for the molecular graph and SMILES string. Two NN encoders try to learn a representation from these correlated views, which feed a shared projection head $g(\cdot)$. The encoders and $g(\cdot)$ are training to maximize representation similarity from the augmented views via a contrastive loss function (eq 9).

1. A traditional practice to acquire positive pairs is to apply data augmentation over a single input representation of the samples in the mini-batch. However, since we are using multiple representations, we defined a family of augmentations for each molecular representation (i.e., node drop, subgraph, edge perturbation, attribute mask, XYZ mask, SMILES enumeration, and SMILES noise). Section 3.2 contains a detailed description of the graph and SMILES transformations. Note that we can simultaneously apply these transformations in both molecular representations.
2. In order to learn a latent representation from the augmentation views, we used a GNN as the base encoder for the molecular graph and an LSTM model for the SMILES. Here, both encoders output a representation with an equal dimension d to avoid using different projection heads. We defined as latent representation the output h_G from the GNN and the h_t from the last layer of the LSTM.
3. The projection head $g(\cdot)$ component maps the latent representation of the previous module into an embedding space z . As stated by Chen et al.,⁵ z led to reduce the loss of information in h_G and h_t influenced by the contrastive loss. The MLP is a common type of NN used as $g(\cdot)$. Thus, our $g(\cdot)$ is an MLP composed of one hidden layer with a rectified linear units (ReLU) activation function and a linear output layer.
4. Finally, a contrastive loss $l(\cdot)$ receives the z representation of h_G and h_t from the samples of the mini-batch. This $l(\cdot)$ aims to change less the NN's weights when the z representations from h_G and h_t of a sample are similar and z from h_G and h_t from different samples are dissimilar. Otherwise, a more significant modification of the NN's weights may occur. This work adopted the normalized temperature-scaled cross-entropy loss (NT-Xent) shown below as $l(\cdot)$.

$$l_{i,j} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)} \quad (9)$$

where $\text{sim}(\mathbf{z}_i, \mathbf{z}_j) = \mathbf{z}_i^T \mathbf{z}_j / \|\mathbf{z}_i\| \|\mathbf{z}_j\|$, and z_i and z_j are a positive pair (i.e., $g(h_G)$ and $g(h_t)$ of the same molecule). Here, $\mathbb{1}_{[k \neq i]} \in \{0,1\}$ is equal to 1 iff $k \neq i$ (i.e., the negative

Table 1. Detailed Description of Augmentations Employed in Molecular Graph and SMILES^a

Data aug.	Description	Assumption	Underlying prior
Molecular graph			
Node drop	Vertices are randomly dropped with their respective edges	Fragment	Vertex missing attempts to make the model learn graph motifs ¹¹
Subgraph	Subgraph extraction using random walk	Fragment	Local information can be a representative description of the molecules ^{11,21}
Edge perturbation	Edges are randomly dropped	Fragment	Link missing attempts to make the model learn the involvements of chemical reactions ²¹
Attribute mask	Vertex's attributes are modified by its respective mean considering all graph nodes	Fragment	Vector corruption attempts to make the model learn chemical information to recover the masked atom given its bonds and neighborhood ^{11,21}
XYZ mask	Provoke a small perturbation in the atoms' coordinates	Ambiguity	Multiple conformers attempt to make the model learn shared aspects among different atom arrangements
SMILES string			
Enumeration	Returns a possible SMILES given an atom as the start point and the approach to traverse the graph	Ambiguity	SMILES enumeration aims to make the model learn relevant aspects shared across multiple string representations ³²
Noise	Replace tokens from the string with the mask token	Fragment	Token corruption attempts to make the model learn chemical information to recover the masked atom and or bond given the context

^aExcept for the SMILES enumeration, all the other augmentation contains a ratio r that controls the level of transformation.

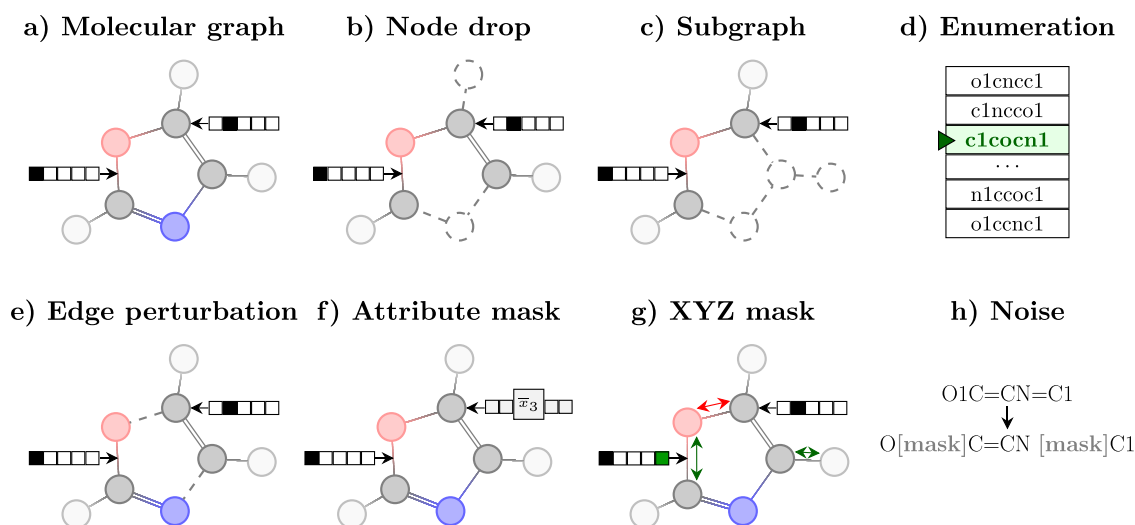


Figure 3. Example of different augmentations in the “O1C=CN=C1” molecule (GDB ID: 53) from the QM9 data set.

pairs), and τ is the temperature parameter. Moreover, since one sample has two augmentation views of itself, the mini-batch contains a total of $2N$ data points, meaning that there are $2(N - 1)$ negative examples for each positive pair.⁵³

3.2. Data Augmentation. In graph-structured data, an appropriate data augmentation may depend on the subject of the graph data set.^{11,12} For instance, a social network data set may be less sensitive to nodes or edges perturbations, whereas these same perturbations may be harmful to molecular graphs since these minor changes can invalidate the compound. Therefore, it is essential to investigate different graph augmentations. In this work, we adopted most of the graph augmentations from the recent works by You et al.¹¹ and Wang et al.²¹ Also, since our framework uses the SMILES string, we proposed a second pool of augmentation specifically for this representation.

Table 1 describes every augmentation used in this work. We categorize them into two assumption groups. The first set of augmentations aims to train a model to learn shared aspects across molecular fragments, which can help the model discover remarkable substructures that characterize the molecule. The second group of augmentation takes advantage of the nonuniqueness of the molecular representations to force the model to learn shared information between different views. For instance, we use the SMILES enumeration to write out possible SMILES string of a molecule, whereas, for the molecular graph, we add a small perturbation in the x , y , and z coordinates to simulate possible molecular nonground-state conformations. Here, we assume that these transformations do not affect the original semantics of the molecular representation. Figure 3 presents an illustration of each transformation.

4. EXPERIMENT CONFIGURATION

In the following, we detail the SMICLR evaluation for semisupervised tasks and unsupervised representation learning.

4.1. Data Sets. In our computational experiments, we explore the QM9³⁴ and MUTAG^{35,36} data sets. QM9 is a popular data set used as a benchmark in several ML algorithms for quantum chemistry applications.^{42,54,55} It comprises 134,000 organic molecules composed of carbon (C), hydrogen (H),

oxygen (O), nitrogen (N), and fluorine (F) with molecules containing at maximum nine heavy atoms. It also reports 15 molecular properties (e.g., geometric, energetic, electronic, and thermodynamic properties) computed with the density functional theory (DFT) B3LYP/6-31G(2df, p) framework. Here, we choose the QM9 to evaluate the SMICLR performance in a semisupervised scenario due to the size of the data set and the challenges in predicting its molecular properties.

MUTAG is another well-known benchmark data set,^{35,36} however, for graph classification applications. It was previously investigated in several studies on representation learning.^{10–12,56} The data set consists of 188 nitroaromatic compounds and an output class that corresponds to the mutagenicity on *Salmonella typhimurium*. Following the recent CL works, we conduct the unsupervised representation learning experiments for a classification problem in the MUTAG data set.¹¹

4.2. Input Representation. Starting from the SMILES string, we construct the molecular graph by defining atoms (chemical bonds) as nodes (edges) of the graph. Then, we describe each node and edge by a feature vector. For the QM9 data set, the feature vector of the nodes is composed of a one-hot encoding of the atom type, the atomic number, whether the atom is aromatic, the atom hybridization (i.e., sp , sp^2 , sp^3), and the number of hydrogen atoms. On the other hand, the feature vector for the edges contains a one-hot encoding of the bond type and the Euclidean distance between the pair of atoms connected by the bond. These are the same features Gilmer et al.¹⁸ used. Here, we adopted the PyTorch Geometric implementation to obtain the molecular graphs,⁵⁷ and all molecular descriptors were computed using the RDKit Python package.⁵¹ For the MUTAG data set, we only represent the nodes by a one-hot encoding of the atom type.

In order to represent the SMILES string for the LSTM, we first define a tokenizer to encode every possible character in the data set as a token. Subsequently, given the SMILES, we obtain a sequence of tokens from the sequence of characters. This sequence of tokens is then fed into an embedding layer, which prompts an embedding vector for each SMILES character. We also padded the SMILES strings to obtain a uniform size according to the string with maximum length. Moreover, we add

an offset of 10 extra characters due to SMILES enumeration, which may produce longer strings than the original ones.

4.3. Model Configuration. For the semisupervised setting, we adopted the enn-s2s model configuration from Gilmer et al.¹⁸ for the GNN base encoder. For the RNN encoder, we used an embedding layer that generates a 32-dimensional embedded representation. This representation is fed directly into an LSTM with a 128-dimensional hidden state. The MLP used as a projection head contains an input layer size equal to 128, a hidden layer with 256 neurons, and an output layer that generates a representation with 384 dimensions. The SMICLR was trained using stochastic gradient descent (SGD) with the Adam optimizer with a learning rate of 1×10^{-3} and weight decay of 1×10^{-6} to optimize the NT-Xent loss with a τ parameter varying from 0.05 to 1. Moreover, we trained the SMICLR with batch sizes equal to 256 and 100 epochs.

After the self-supervised phase, an MLP (σ) is stacked on the SMICLR's base encoder, as shown in Figure 4. This MLP

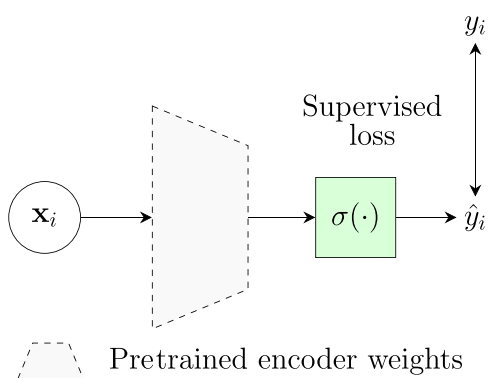


Figure 4. Given a $\langle x_i, y_i \rangle$ pair, where x_i is the molecular representation and y_i is the molecular property, the supervised model will try to predict y_i using a base encoder from SMICLR and an MLP. Initially, the base encoder receives the pretrained weights from the self-supervised phase. Then, all parameters of this NN are adapted to predict y .

contains one hidden layer with 64 hidden units followed by the ReLU activation function. Then, this NN is trained to fine-tune the encoder's parameters and predict the desired property. In this supervised phase, we adopted the exact SGD before to optimize the mean squared error and used a batch size and number of epochs equal to 32 and 300, respectively. Here, data augmentation is not applied. Furthermore, since SMICLR has two encoders, we tested which learned representation improves our results.

For the unsupervised task, instead of using the enn-s2s as the GNN-base encoder, we employed the GIN as in previous studies.^{10,11} In our experiment, we changed the learning rate to between 1×10^{-2} and 1×10^{-3} , the batch sizes between 32 and 128, number of GNN layers from 2 to 3, under different data augmentation and r changing between 0.1 and 0.2. We used PyTorch and PyTorch Geometric packages to implement the models.⁵⁸

4.4. Training Details. We conduct the same semisupervised procedure as Sun et al.¹⁰ First, we remove from the QM9 data set the 3054 molecules that failed the geometry consistency check.³⁴ Next, we normalize all properties to have zero mean and unit variance. Subsequently, we divided the QM9 into a training set with 110,831 samples and a validation/test set with 10,000 samples each. Then, we randomly labeled a subset of 5000 samples from the training set. In the SSL training, we used

the same augmentation ratio parametrized by You et al.,¹¹ which has the value of 0.2. In order to compare our results, we report the performance of the state-of-the-art method on molecular prediction property: the InfoGraph* and its mean-teacher adaptation.¹⁰ For the InfoGraph*, we adopted the same SGD optimizer, batch size, and split in the QM9 described before. We also compare our results with a purely supervised baseline version of our model.

For the unsupervised representation learning, we followed the same protocols adopted by You et al.¹¹ and Sun et al.,¹⁰ which evaluate the representation learned by the base encoder in a downstream support vector machine (SVM) classifier. Then, our results are compared with three state-of-the-art graph kernel methods, i.e., graphlet kernel (GL),⁵⁹ Weisfeiler-Lehman subtree kernel (WL),⁶⁰ deep graph kernel (DGK),⁶¹ and six unsupervised graph-level representation learning methods, i.e., node2vec,⁶² sub2vec,⁶³ graph2vec,⁵⁶ InfoGraph,¹⁰ GraphCL,¹¹ and the Hassani's CL model.¹²

5. RESULTS AND DISCUSSION

This section discusses the results obtained in our computational experiments. We start by providing an investigation of the latent space learned by our framework. Then, we study the impact of data augmentation and evaluate the SSL results on predicting 12 molecular properties of the QM9 data set. Here, the SSL results are in terms of mean absolute error (MAE). Note that we adopted several hyperparameters from previous works, as mentioned in Section 4, and consolidated our search for the best model by investigating data augmentations and other critical design choices.

Moreover, due to computational constraints, we limited this hyperparameter tuning toward the prediction of atomization energy at 0 K (U_0) and assessed the results from the tuned framework in the remaining molecular properties. We also assess the unsupervised representation learning in a graph classification task using the MUTAG data set. Furthermore, we computed the MAE using the validation set when performing the hyperparameter search. Then, we used the test set to report the model's performance with tuned hyperparameters.

5.1. Qualitative Analysis of the Latent Representation.

Herein, we investigate the latent space learned by SMICLR to understand what molecular characteristics were relevant for our method. To this end, we first use the t-distributed stochastic neighbor embedding (t-SNE) to embed the 128-dimensional h_G representation into a bidimensional space. The choice of h_G relied on its better capacity to predict the U_0 , as we see in the subsequent section. t-SNE is a nonlinear dimensionality reduction algorithm that tries to preserve similar data points in the original representation by mapping them as adjacent points in the low-dimensional space. Figure 5 shows the t-SNE embedding of the latent space for the validation set.

Our investigation started by looking at molecular patterns related to topological descriptors. As a result, we divided the latent space into three categories: (i) the acyclic molecules, which stand for molecules without rings, (ii) the molecular group that contains rings that do not share atoms between them, and (iii) the class of molecules with rings that share two or more atoms, namely, fused rings. We also observed a fourth group; however, in this case, the structure characteristic is related to molecules composed of F atoms.

Moreover, since there is no single feature capable of capturing the variation of the molecules, we decided to explore each of the before-mentioned groups, trying to find molecular properties

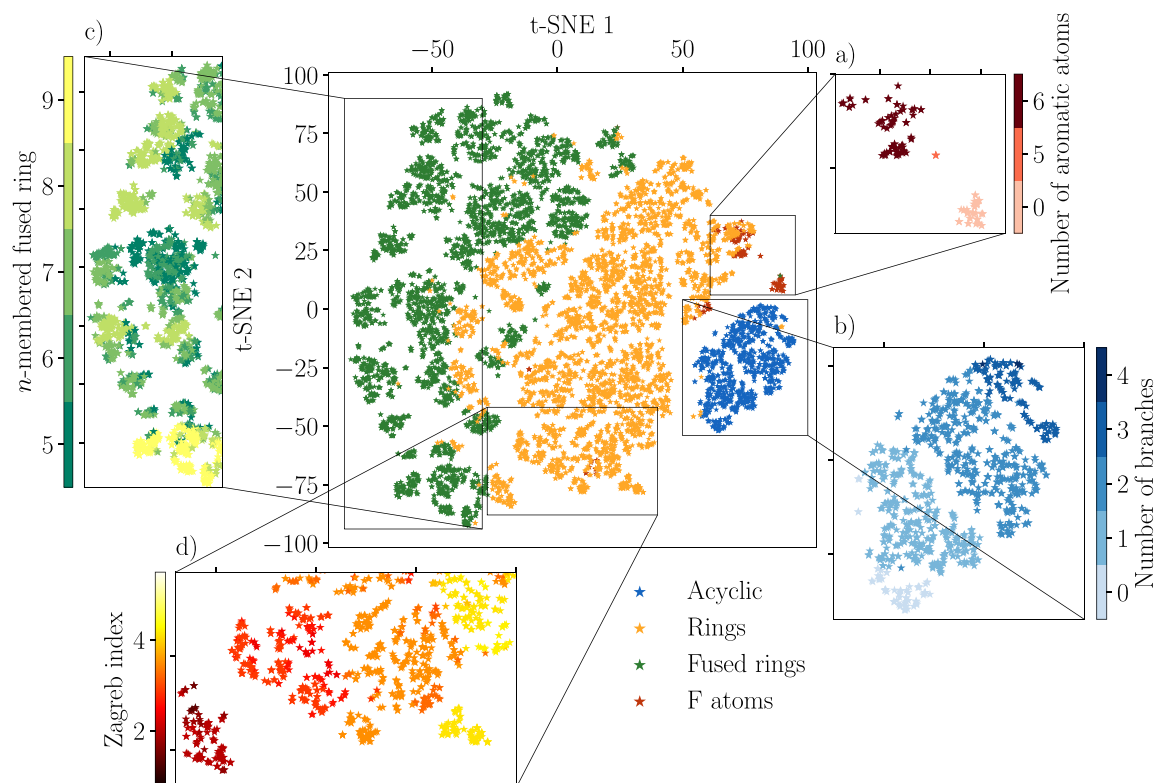


Figure 5. t-SNE low-dimensional representation of the h_G latent space learned by SMICLR of the validation set.

that may describe subclusters. Looking carefully at the acyclic molecules, we discover that this cluster can be split into at least five other subgroups when considering the number of branches in the compound structure, as shown in Figure 5b. Table 2 gives an example of a molecule in each subgroup. A branch in SMILES representation is denoted by enclosing a substructure in parentheses.

Table 2. Example of Molecules with Different Numbers of Branches (# br.)^a

# br.	SMILES	GDB ID
0	OCCCCOCC#N	123215
1	CC#CC(=NO)C#C	4001
2	CC(CO)(CC=O)C#N	100363
3	CC(C)C(C#C)C(C)C	58034
4	CC(C)C(C(C)C(N)=O	58086

^aThese molecules were extracted from the acyclic group.

Concerning the molecules with the halogen atom, F, we noted that the number of aromatic atoms could further arrange its group into three small subgroups, as illustrated in Figure 5a. For the molecular group with fused rings, we observed the presence of subgroups in relation to the number of atoms that compose the fused ring, as shown in Figure 5c. However, it is possible to note that the same subgroup may also be spread out at different locations of the latent space. Thus, this example illustrates that the latent representation encompasses diverse information about the molecules, making molecules with different numbers of atoms in the ring more similar than molecules with the same number of members.

Finally, for the rings group (Figure 5d), we were capable of distinguishing its molecules by looking at the Zagreb index. The

Zagreb index is a topological molecular descriptor that describes the molecule based on the vertex degree of its atoms, as defined below⁶⁴

$$\lambda_{M_1} = \sum_{i=1}^n d_i^\lambda \quad (10)$$

where d is the degree of the i th atom, n the number of atoms, and λ a real number, which here is equal to -1 . Therefore, low λ_{M_1} indicates molecules with a high degree vertex. Similarly, high λ_{M_1} values indicate molecules with low degree values.

5.2. Choice of the Base Encoder. Since the SMICLR framework comprises two base encoders, we investigated which learned representation contributes utmost to improve our model's predictive performance. For this, we evaluated the latent representation from each base encoder individually following the processes described in Section 4.3. Then, we assess the performance when concatenating the representation of both base encoders. Each result was computed based on the validation error, in terms of MAE, of five executions from scratch when predicting the U_0 and without augmentations in the CL process. As a result, the enn-s2s and LSTM model showed errors of 0.157 ± 0.007 and 0.236 ± 0.004 eV, respectively, whereas when using the representation of both encoders, we achieve an error of 0.169 ± 0.005 eV. This find shows a good agreement with what is observed in the literature,^{18,42} in which the GNN performs better in predicting molecular properties. Moreover, this result exhibits a reduction of $\sim 25\%$ over the purely supervised GNN error, which achieved an MAE of 0.1957 eV. For this reason, we adopted the GNN encoder in the subsequent experiments to predict the molecular properties.

Table 3. Validation Error for Semisupervised Training on Compositions of Augmentations to Predict U_0^a

SMILES \ Graph	Identical	Node drop	Subgraph	Edge pert.	Attr. mask	XYZ mask
Identical	0.8	0.56	1.275	0.779	0.695	0.764
Enumeration	0.76	0.559	1.176	0.797	0.657	0.749
Noise	0.738	0.512	1.259	0.691	0.601	0.75

^aRows show transformation on the SMILES string, while the columns are the molecular graph. Each cell reports the error ratio of our semisupervised model with respect to the purely supervised version of the enn-s2s. Values below (above) 1 mean better (worse) performance than this supervised baseline. This supervised baseline error, in terms of MAE, is equal to 0.1957 eV. We compute these results over five runs from scratch.

5.3. Improving Accuracy with Data Augmentation.

Several works have shown that data augmentation for CL can improve accuracy. Hence, this became a crucial step in this setting. To investigate the impacts of seven data augmentations, we first applied an augmentation for only one molecular representation and maintained the other identical. Next, we made all possible combinations constrained by one augmentation for each representation. Table 3 shows the validation set performance, under the protocol described in Section 4.4, of the SMICLR on all the different compositions of augmentations.

The first notable result in Table 3 is that most of the experimental cases surpass the supervised baseline, even when using the raw molecular representations (the identical–identical cell), which by themselves are two viewpoints of the same molecule, i.e., showing that our CL framework benefits from different modalities. Moreover, our best performance was achieved by the composition of multiple augmentations. Another remark is that graph transformations tend to reduce more the supervised error than the SMILES augmentations. Specifically, graph manipulations that affected the atoms (e.g., node drop, attribute mask) demonstrated more accurate models than edge modifications (e.g., edge perturbation, XYZ mask). Regarding this observation, we speculate the following.

The bond information seems to be more sensitive in graph contrastive tasks. As reported by You et al.,¹¹ even though edge perturbation increased accuracy for the biomolecule data PROTEINS,^{65,66} it also hurt performance for the biomolecule data NCI1. This sensitivity may be tied to the fact that a change in the bond may invalidate the entire compound or modify its identity. While here edge perturbation did not degrade the accuracy, our model seems to map the same molecule into the exact point of the feature space no matter what modifications were caused by this augmentation. In contrast, the latent space obtained using augmentations that modify the molecule's atoms appears to map the different augmented views at distinct but near points (see the Supporting Information for more details). Therefore, edge augmentations may have influenced less the downstream task due to the loss of information.

Furthermore, the subgraph augmentation deteriorates the performance differently from the results presented by You et al.¹¹ on other molecular data sets. We argue that the QM9's molecules may be sensitive to changes in the graph topology because it is composed of small molecules. For instance, the average number of atoms in the QM9's molecules is 18, whereas the NCI1 and PROTEINS data set used by You et al.¹¹ contain molecules with 29 and 39 atoms on average, respectively. In addition, since this transformation removes nodes and edges to generate a subgraph with the number of vertices concerning the

augmentation ratio, it may oversimplify the molecular graph. Thus, perform poorly in predicting the molecular property.

On the other hand, in the SMILES augmentation, we observed that results tend to be better when adding noise to the string. Figure 6 shows the contrastive loss when combining

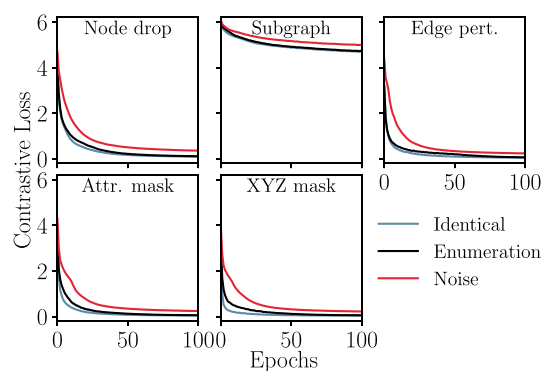


Figure 6. Training contrastive loss behavior through the epochs. Boxes represent the loss when fixing a graph augmentation and varying the SMILES transformations (lines). The lines express the mean of five training runs.

different data augmentation. When looking at the learning curve obtained by the SMILES enumeration, it is possible to note that the error decays fast with a behavior almost similar to the curve when not using an augmentation. In contrast, the loss curve when using the SMILES noise seems to make the CL harder, which probably made our model avoid learning trivial features and improve the quality of the representation. Also, in Figure 6, it is possible to observe how the model struggles to learn when using the subgraph augmentation, thus resulting in poor predictive performance. In summary, our best result was achieved using the node drop augmentation in the molecular graph and the noise in the SMILES string.

5.4. Adjusting the Temperature. We evaluated the XENT-loss varying the τ hyperparameter to find an appropriate τ value that helps the model to distinguish better positive and negative samples. For this investigation, we explore common τ values founded in the literature: 0.05, 0.1, 0.5, and 1. Table 4 reports the validation error over five executions of our SSL model from scratch. Our best results were achieved using $\tau = 0.1$ followed by 0.05, 0.5, and 1. This find demonstrates consistency with previous results presented in the literature for molecular data application.²¹

5.5. Semisupervised and Unsupervised Learning. In this section, we first evaluated the SMICLR's performance when predicting energetic, electronic, and thermodynamic properties

Table 4. Impact of Different τ Values When Predicting the Atomization Energy at 0 K^a

τ	MAE (eV)
0.05	0.1077 $\pm$ 0.0008
0.10	0.1002 $\pm$ 0.0025
0.50	0.1129 $\pm$ 0.0037
1.00	0.1173 $\pm$ 0.0031

^aWe compute these results over five runs from scratch. Reported errors were obtained using the validation set.

from the QM9 data set in an SSL scenario. For this experiment, we considered the combination of data augmentation that delivered the best result in our framework, i.e., the node drop for the molecular graph and adding noise in the SMILES, as shown in Table 3. Table 5 presents the test error, in terms of MAE, for 12 molecular properties. We compare these results with the supervised baseline, InfoGraph* model,¹⁰ and SMICLR framework without data augmentation. In general, we observed that all SSL methods show an accuracy improvement concerning the purely supervised baseline, demonstrating that adding unlabeled data in the training process helps achieve better results. Moreover, our framework significantly reduced the predictive error for all of the 12 properties.

Nevertheless, it is noteworthy that the graph-structured representation required by the enn-s2s model uses atomic distances to describe the molecular configuration. In our initial experimental setup, we computed these distances from the DFT-optimized geometries reported in the QM9 data set, similar to the prior work developed by Sun and collaborators.¹⁰ However, DFT calculations demand a lot of computational resources, which can be a drawback for practical applications.⁶⁷ Therefore, we evaluated our model performance using geometries optimized at a lower level of theory.

For this purpose, we followed the same protocol adopted to build Table 5, however, using molecular conformations calculated via the MMFF94 force field for the self-supervised training stage. Here, we used make3D and localopt functions from the OpenBabel Python package (version 3.1.0) to generate the 3D coordinates given the SMILES string and to run the

MMFF94 force field optimization, respectively.⁵² In order to assess the sensitivity of our model over initial conditions of the force field method, we used molecular geometries obtained from different MMFF94 calculations for each one of the five executions. Table 6 reports the test set performances conducted in this battery of tests. While not achieving the best result, it also does not hurt the performance compared with the SMICLR without data augmentation trained on DFT-optimized geometries.

Moreover, the errors obtained with the force field geometries in Table 6 agree with our previous result when using the XYZ mask augmentation, where the rearrangement of the atom's positions did not degrade our model performance. Furthermore, these results indicate that CL may take advantage of other state-of-the-art GNN models that use geometric information, such as SchNet.²⁰

The second part of this section evaluates the SMICLR in unsupervised representation learning. Table 7 shows the accuracy of the SMICLR representation for the classification problem of the MUTAG data set. It is noted that our method outperformed the other ML models and exhibited the lowest error standard deviation.

6. INSIGHTS ON THE CONTRASTIVE LEARNING PROCESS

Neural models trained to learn generalizable feature representations from unlabeled data can help improve predictive accuracy. Here, our contrastive framework played such a role when combining different data modalities and augmentation techniques. While the purpose of using multiple modalities was to learn shared information between different ways of describing a molecule, the objective of the augmentation methods depends on its transformations. For instance, the idea of using augmentation that extracts molecular substructures was to learn shared aspects across such fragments. On the other hand, transformations that explore the ambiguity of a molecular representation have a similar goal of using different modalities.

Interestingly, we achieved the best performance using augmentations that generate fragments. We hypothesize that combining these augmentations with modalities brings more

Table 5. Test Error for Semisupervised Predictions of the Following Molecular Properties of the QM9 Data Set: (i) Dipole Moment (μ), (ii) Isotropic Polarizability (α), (iii) Highest Occupied Molecular Orbital Energy (ϵ_{HOMO}), (iv) Lowest Unoccupied Molecular Orbital Energy (ϵ_{LUMO}), (v) Energy Band Gap between the ϵ_{HOMO} and ϵ_{LUMO} States ($\Delta\epsilon$), (vi) Electronic Spatial Extent ($\langle R^2 \rangle$), (vii) Zero Point Vibrational Energy (ZPVE), (viii) Heat Capacity at 298.15 K (c_v), (ix) Atomization Energy at 298.15 K (U), (x) Atomization Enthalpy at 298.15 K (H), and (xi) Atomization Free Energy at 298.15 K (G)^a

Target	Supervised	InfoGraph* ¹⁰	Mean Teacher ¹⁰	SMICLR	
				w/o aug.	w/aug.
μ (D)	0.22 $\pm$ 0.004	0.218 $\pm$ 0.003	0.222 $\pm$ 0.002	0.218 $\pm$ 0.006	0.185 $\pm$ 0.009
α (a03)	0.484 $\pm$ 0.007	0.499 $\pm$ 0.006	0.499 $\pm$ 0.005	0.436 $\pm$ 0.025	0.36 $\pm$ 0.005
ϵ_{HOMO} (eV)	0.155 $\pm$ 0.001	0.161 $\pm$ 0.003	0.153 $\pm$ 0.001	0.144 $\pm$ 0.004	0.132 $\pm$ 0.002
ϵ_{LUMO} (eV)	0.15 $\pm$ 0.003	0.148 $\pm$ 0.002	0.144 $\pm$ 0.001	0.136 $\pm$ 0.002	0.12 $\pm$ 0.001
$\Delta\epsilon$ (eV)	0.228 $\pm$ 0.002	0.235 $\pm$ 0.002	0.22 $\pm$ 0.002	0.217 $\pm$ 0.003	0.191 $\pm$ 0.003
$\langle R^2 \rangle$ (a0 ²)	2.928 $\pm$ 0.147	2.71 $\pm$ 0.27	1.45 $\pm$ 0.087	1.878 $\pm$ 0.192	1.329 $\pm$ 0.096
ZPVE (eV)	0.009 $\pm$ 0	0.009 $\pm$ 0	0.007 $\pm$ 0	0.007 $\pm$ 0	0.006 $\pm$ 0
c_v (cal/mol K)	0.183 $\pm$ 0.009	0.172 $\pm$ 0.003	0.165 $\pm$ 0.002	0.145 $\pm$ 0.006	0.118 $\pm$ 0.004
U_0 (eV)	0.195 $\pm$ 0.005	0.187 $\pm$ 0.008	0.146 $\pm$ 0.003	0.157 $\pm$ 0.008	0.1 $\pm$ 0.002
U (eV)	0.194 $\pm$ 0.003	0.195 $\pm$ 0.007	0.149 $\pm$ 0.006	0.159 $\pm$ 0.003	0.103 $\pm$ 0.001
H (eV)	0.198 $\pm$ 0.006	0.19 $\pm$ 0.01	0.14 $\pm$ 0.002	0.158 $\pm$ 0.002	0.104 $\pm$ 0.005
G (eV)	0.196 $\pm$ 0.006	0.186 $\pm$ 0.011	0.136 $\pm$ 0.007	0.15 $\pm$ 0.005	0.098 $\pm$ 0.003

^aResults in bold indicate best performances.

Table 6. Test Error When Using XYZ Obtained from the MMFF94 Force Field in the Self-Supervised Phase^a

Target	SMICLR (DFT)		SMICLR (MMFF94)	
	w/o aug.	w/aug.	w/o aug.	w/aug.
μ (D)	0.218 $\pm$ 0.006	0.185 $\pm$ 0.009	0.216 $\pm$ 0.006	0.194 $\pm$ 0.006
α (a03)	0.436 $\pm$ 0.025	0.36 $\pm$ 0.005	0.442 $\pm$ 0.007	0.388 $\pm$ 0.009
ϵ_{HOMO} (eV)	0.144 $\pm$ 0.004	0.132 $\pm$ 0.002	0.145 $\pm$ 0.003	0.131 $\pm$ 0.002
ϵ_{LUMO} (eV)	0.136 $\pm$ 0.002	0.12 $\pm$ 0.001	0.142 $\pm$ 0.003	0.124 $\pm$ 0.001
$\Delta\epsilon$ (eV)	0.217 $\pm$ 0.003	0.191 $\pm$ 0.003	0.224 $\pm$ 0.003	0.196 $\pm$ 0.004
$\langle R^2 \rangle$ (a0 ²)	1.878 $\pm$ 0.192	1.329 $\pm$ 0.096	2.4 $\pm$ 0.235	1.428 $\pm$ 0.038
ZPVE (eV)	0.007 $\pm$ 0.0	0.006 $\pm$ 0.0	0.008 $\pm$ 0.0	0.006 $\pm$ 0.0
c_p (cal/mol K)	0.145 $\pm$ 0.006	0.118 $\pm$ 0.004	0.152 $\pm$ 0.002	0.128 $\pm$ 0.003
U_0 (eV)	0.157 $\pm$ 0.008	0.1 $\pm$ 0.002	0.169 $\pm$ 0.01	0.11 $\pm$ 0.003
U (eV)	0.159 $\pm$ 0.003	0.103 $\pm$ 0.001	0.168 $\pm$ 0.009	0.114 $\pm$ 0.002
H (eV)	0.158 $\pm$ 0.002	0.104 $\pm$ 0.005	0.171 $\pm$ 0.006	0.112 $\pm$ 0.003
G (eV)	0.15 $\pm$ 0.005	0.098 $\pm$ 0.003	0.165 $\pm$ 0.009	0.11 $\pm$ 0.005

^aResults in bold indicate best performances.Table 7. Graph Classification Accuracy of the Test Set When Using the Representation Learned by SMICLR into an SVM Classifier^a

Model	Accuracy (%)	Ref
GL ⁵⁹	81.66 $\pm$ 2.11	11
WL ⁶⁰	80.72 $\pm$ 3.00	10
DGK ⁶¹	87.44 $\pm$ 2.72	10
node2vec ⁶²	72.63 $\pm$ 10.20	10
sub2vec ⁶³	61.05 $\pm$ 15.80	10
graph2vec ⁵⁶	83.15 $\pm$ 9.25	10
GraphCL ¹¹	86.80 $\pm$ 1.34	11
InfoGraph ¹⁰	89.01 $\pm$ 1.13	10
Hassani ¹²	89.70 $\pm$ 1.1	12
Ours	89.98 $\pm$ 0.51	

^aResults in bold indicate the best performance. The last column contains the reference to the reported result.

variation in the input representation than combining the augmentations with the same purpose of using different modalities. However, using the subgraph transformation led to a deterioration of the model accuracy. Since this manipulation can be seen as the combination of node drop and edge perturbation, we conjecture that it removed too much information from the molecules of the QM9 data set.

Moreover, the embedding space learned by our model illustrated in Figure 5 allows us to note how the model distributed the molecules in the space. It is possible to note that molecules with similar (distinct) characteristics are close (far away) in the latent space. For instance, we identified groups of compounds composed of at least one fluorine atom, acyclic molecules, or molecules with fused and nonfused rings. When looking carefully, it can be observed that the fused ring group is closer to the cluster of molecules with rings that are not fused than the group with acyclic molecules. Finally, Figure 5 also shows how the SMILES string influences the latent representation learned by the enn-s2s encoder. For example, the group of acyclic molecules can be further divided when considering only explicit patterns in the SMILES, such as the ramifications described in the string.

7. CONCLUSION

We proposed a self-supervised framework, called SMICLR, that learns molecular representation. For this, we jointly trained a GNN and LSTM encoder to maximize agreement between

correlated views generated from molecular modalities of the same molecule (i.e., the molecular graph and SMILES string) and data augmentation techniques. Then, once SMICLR was trained, we used its representations to improve predictive molecular tasks. In our experiments, we investigated the benefit of using data augmentation methods, loss function configuration, and the impact of the base encoder in the final supervised task. By combining our findings, SMICLR surpasses other literature methods in predictive tasks in the semi-supervised and unsupervised representation learning scenario.

Specifically, when training SMICLR to predict the QM9 properties under a semisupervised protocol, our model showed the lowest error for all different properties. Our best result was achieved when predicting the $\langle R^2 \rangle$, in which we reduced the error by roughly 50% over the purely supervised baseline. We also achieved a similar error reduction when predicting the following properties: U_0 , U , H , and G . On the other hand, our worst case was predicting the ϵ_{HOMO} , ϵ_{LUMO} , and $\Delta\epsilon$, when we reduced the error by almost 15% over the purely supervised baseline, which still shows an improvement. Moreover, our approach shows competitive results in unsupervised representation learning by achieving the best result accuracy in the graph classification problem of the MUTAG data set.

In addition, our results reveal opportunities to improve predictive tasks by incorporating effectively unlabeled data from the chemical space and depending less on labeled data. These findings demonstrate possibilities to achieve better results without performing additional calculations to obtain quantum properties, which can demand a lot of computational resources.

Nevertheless, our method currently has noteworthy limitations. First, the hyperparameter tuning on the QM9 data set was biased toward one particular molecular property, U_0 . By doing this, we ensure the best result for U_0 , while it does not guarantee the same performance for other molecular properties. Therefore, a future investigation will explore design choices that further improve our framework for each target property. In the Supporting Information, we provide an initial study on the impact of composing augmentations for ϵ_{HOMO} . Second, our SSL study concentrates on the QM9 data set, which has some drawbacks regarding chemical diversity. This data set was built on the GDB database with restrictions concerning small molecules with up to 29 atoms drawn from the elements H, C, N, O, and F. Hence, exploring more data sets to cover the chemical space diversity may improve representation learning

and generalization. This observation is also valid for other downstream applications.

In future work, we intend to investigate the before-mentioned limitations. We also intend to understand more the relationship between the features learned by our model and molecular descriptors. Moreover, we plan to extend our framework for multiple molecular tasks, for instance, through the pretraining model, coupling the base encoder of the SMILES in a model that performs molecular *de novo* design while using the graph base encoder in models that predicts molecular properties.

8. DATA AND SOFTWARE AVAILABILITY

For the sake of reproducibility, we release the source code of the SMICLR model in a public GitHub repository: <https://github.com/CIDAG/SMICLR>. The QM9 data set is available in the PyTorch Geometric package. The graph-structured data of the MUTAG data set is available on <https://chrsmrrs.github.io/datasets/docs/datasets/> and its SMILES string version on https://github.com/PaddlePaddle/PaddleHelix/blob/dev/apps/pretrained_compound/info_graph/README.md. The chemoinformatic packages (RDKit and OpenBabel) are available at <https://www.rdkit.org/> and http://openbabel.org/wiki/Main_Page. The PyTorch Geometric package is available at https://github.com/pyg-team/pytorch_geometric.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.2c00521>.

Data employed for the figures, as well as complementary analyses and additional technical details (PDF)

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Notes

The authors declare no competing financial interest.

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